

MongoDB Aggregations med flera tabeller - övningar

Precis som med SQL JOINS övningarna,
Skapa en ny tabell departments som nedan

departments

id	department_name
1	Tech support
2	Marketing
3	Sales
4	Department with no employees

Skapa en relation mellan collections departments och employees. Detta görs genom att införa en ny kolumn i employees med kolumnnamnet departments_id. departments_id refererar till id från departments. Använd updateMany() för att lägga till departments_id för för samtliga anställda, förutom "Kalle Anka"

Obs! Denna övning bygger vidare på din employees tabell från tidigare MongoDB övningar.

employees

id	fname	lname	salary	employed_date	department_id
Fylls i automatiskt	Göran	Dahl	34700	Fylles i automatiskt	1
xx	Eva	Dahl	33000	xxxx-xx-xx xx:xx:xx	3
xx	Robin	Hood	43500	xxxx-xx-xx xx:xx:xx	1
xx	Kalle	Anka	55000	xxxx-xx-xx xx:xx:xx	
xx	Göran	Persson	55000	xxxx-xx-xx xx:xx:xx	2
xx	Gunnel	Eriksson	39000	xxxx-xx-xx xx:xx:xx	3
xx	Hans	Svärd	44400	xxxx-xx-xx xx:xx:xx	3
xx	Anna	Svärd	54400	xxxx-xx-xx xx:xx:xx	3

```
db.departments.insertMany([
  { department_name: "Tech support"},
  { department_name: "Marketing"},
  { department_name: "Sales"},
  { department_name: "Department with no employees"}
]);
```

```
db.employees.updateOne(
  { fname: "Göran", lname: "Dahl" },
  { $set: { department_id: ObjectId("xxx") }}
);
```

```
db.employees.updateOne(
  { fname: "Eva", lname: "Dahl" },
  { $set: { department_id: ObjectId("xxx") }}
);
```

```
db.employees.updateOne(
  { fname: "Robin", lname: "Hood" },
  { $set: { department_id: ObjectId("xxx") }}
);
```

```
db.employees.updateOne(
  { fname: "Kalle", lname: "Anka" },
  { $set: { department_id: null }}
);
```

```
db.employees.updateOne(
  { fname: "Göran", lname: "Persson" },
  { $set: { department_id: ObjectId("xxx") }}
);
```

```

db.employees.updateOne(
  { fname: "Gunnel", lname: "Eriksson" },
  { $set: { department_id: ObjectId("xxx") } }
);
db.employees.updateOne(
  { fname: "Hans", lname: "Svärd" },
  { $set: { department_id: ObjectId("xxx") } }
);
db.employees.updateOne(
  { fname: "Anna", lname: "Svärd" },
  { $set: { department_id: ObjectId("xxx") } }
);

```

Nu är det dags för aggregate övningarna

1. Visa alla employees som är i Tech support

```

db.employees.aggregate([
  {$lookup: {
    from: "departments",
    localField: "department_id",
    foreignField: "_id",
    as: "departments_details",
  }},

  { $unwind: "$departments_details" },
  { $match: { "departments_details.department_name": "Tech support" } },

  {$project: {
    _id: 0,
    fname: 1,
    lname: 1,
    department: "$departments_details.department_name"
  }}
])

```

2. Visa alla employees som är i Marketing och Sales

```
db.employees.aggregate([
  {$lookup: {
    from: "departments",
    localField: "department_id",
    foreignField: "_id",
    as: "departments_details",
  }},
  { $unwind: "$departments_details" },

  {$match: {"$or": [
    {"departments_details.department_name": "Marketing"} ,
    {"departments_details.department_name": "Sales"}
  ]}
},

{$project: {
  _id: 0,
  fname: 1,
  lname: 1,
  department: "$departments_details.department_name"
}}
])
```